

CHUGUYEVKA, Russian Federation

WMO: 319390

Lat: 44.150N

Lon: 133.867E

Elev: 259

StdP: 98.25

Time Zone: 10.00 (E10)

Period: 90-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.2	-30.8	-35.5	0.1	-32.9	-32.9	0.2	-30.5	5.5	-14.4	5.0	-14.7	0.5	120	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	30.2	21.9	28.6	20.9	27.2	20.2	23.7	28.2	22.5	26.7	21.4	25.4	1.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.5	26.5	21.1	16.3	25.0	20.1	15.2	23.8	72.6	27.9	68.0	26.6	63.9	25.5	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.4	4.5	3.8	-35.5	33.1	3.0	1.6	-37.7	34.2	-39.4	35.1	-41.1	36.0	-43.3	37.2
			-35.6	24.9	3.0	1.7	-37.7	26.2	-39.5	27.2	-41.1	28.2	-43.3	29.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-19.8	-13.3	-4.4	4.6	12.1	17.3	21.3	20.3	13.8	5.3	-5.5	-16.1	(6)	
(7)		DBStd	14.62	4.64	5.40	5.58	3.75	3.66	2.98	2.70	2.91	3.42	4.46	6.14	6.09	(7)	
(8)		HDD10.0	3645	922	653	448	166	21	1	0	0	7	157	463	808	(8)	
(9)		HDD18.3	5777	1181	887	706	413	193	53	6	14	141	404	713	1066	(9)	
(10)		CDD10.0	1111	0	0	0	3	87	221	350	319	121	10	0	0	(10)	
(11)		CDD18.3	202	0	0	0	0	2	23	97	74	5	0	0	0	(11)	
(12)		CDH23.3	1657	0	0	0	2	69	268	779	501	36	1	0	0	(12)	
(13)		CDH26.7	424	0	0	0	0	13	54	232	120	5	0	0	0	(13)	
(14)	Wind	WSAvg	1.3	0.9	1.3	1.6	1.9	1.7	1.3	1.3	1.1	1.1	1.4	1.5	1.0	(14)	
(15)	Precipitation	PrecAvg	620	13	12	23	35	64	67	109	101	81	56	34	21	(15)	
(16)		PrecMax	937	26	34	72	72	128	145	226	219	195	127	67	50	(16)	
(17)		PrecMin	289	1	1	2	6	6	16	42	26	16	8	4	2	(17)	
(18)		PrecStd	168	6	8	16	19	30	30	53	52	51	26	16	12	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.8	4.8	13.3	22.3	29.2	30.2	32.7	32.1	27.3	21.5	12.2	2.8	(19)	
(20)			MCWB	-5.1	1.2	6.1	11.3	15.8	19.2	24.0	23.4	20.7	13.9	7.4	1.2	(20)	
(21)		2%	DB	-6.7	1.2	9.9	19.1	25.6	28.2	30.9	29.5	23.8	18.0	8.6	0.3	(21)	
(22)			MCWB	-7.9	-1.3	4.2	9.4	15.2	18.9	23.0	22.3	17.7	11.8	5.6	-0.8	(22)	
(23)		5%	DB	-9.4	-1.4	6.8	15.4	23.2	26.6	29.1	27.7	21.9	15.5	5.8	-3.8	(23)	
(24)			MCWB	-10.3	-3.2	2.2	7.8	13.3	18.3	21.8	21.5	16.7	10.4	3.2	-4.7	(24)	
(25)		10%	DB	-11.5	-3.7	4.5	13.0	21.0	24.5	27.6	26.1	20.3	13.2	3.2	-6.1	(25)	
(26)			MCWB	-12.3	-5.3	0.6	6.5	12.7	17.8	21.2	20.8	15.8	9.1	1.0	-7.1	(26)	
(27)		Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.7	1.3	6.7	11.9	17.4	21.4	25.4	25.0	21.8	14.8	8.3	1.8	(27)
(28)				MCDB	-3.7	3.9	11.5	21.2	25.4	27.1	31.0	30.7	26.4	20.7	11.0	2.8	(28)
(29)	2%		WB	-7.8	-0.9	4.3	9.8	15.9	19.8	24.1	23.5	19.1	12.6	5.8	-1.3	(29)	
(30)			MCDB	-6.9	0.9	9.1	17.3	23.6	26.3	29.1	27.6	22.0	16.5	8.3	0.1	(30)	
(31)	5%		WB	-10.3	-3.1	2.4	8.3	14.5	18.9	23.0	22.4	17.7	11.1	3.6	-4.5	(31)	
(32)			MCDB	-9.5	-1.5	6.5	14.8	20.8	25.0	27.5	26.3	20.6	14.7	5.8	-3.8	(32)	
(33)	10%		WB	-12.3	-5.4	0.9	6.8	13.3	18.1	21.9	21.4	16.5	9.5	1.2	-7.2	(33)	
(34)			MCDB	-11.5	-3.7	3.9	12.2	19.4	23.9	26.2	25.0	19.5	12.7	2.8	-6.2	(34)	
(35)	Mean Daily Temperature Range		5% DB	MDBR	14.8	14.7	12.6	11.9	12.4	10.7	10.1	9.9	11.1	12.2	11.1	12.3	(35)
(36)				MCDBR	16.0	15.1	15.0	18.0	17.9	15.3	13.5	12.6	13.7	16.4	13.9	11.9	(36)
(37)		MCWBR		14.7	12.8	9.8	10.3	8.4	6.9	6.0	6.2	7.5	10.0	10.3	10.6	(37)	
(38)		5% WB	MCDBR	15.7	14.8	13.8	17.1	15.2	13.6	12.0	11.0	10.5	14.7	12.8	11.4	(38)	
(39)			MCWBR	14.4	12.5	9.2	10.3	7.8	6.7	5.9	5.9	6.2	9.4	9.8	10.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.317	0.381	0.463	0.561	0.567	0.540	0.501	0.465	0.422	0.433	0.382	0.338	(40)		
(41)		taud	2.056	1.877	1.772	1.692	1.748	1.902	2.078	2.181	2.214	2.033	2.056	2.019	(41)		
(42)		Ebn at Noon	766	766	750	713	727	750	776	792	795	708	677	686	(42)		
(43)		Edh at Noon	104	152	193	228	222	191	158	138	123	129	106	99	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2.14	3.13	4.11	4.56	5.05	5.39	4.87	4.56	4.08	2.96	2.02	1.71	(44)		
(45)		RadStd	0.14	0.17	0.29	0.43	0.43	0.58	0.51	0.40	0.34	0.28	0.19	0.11	(45)		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page